

Bank Runs, Deposit Insurance, and Liquidity

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ABSTRACT

This paper develops a model of banking grounded in the provision of liquidity insurance rather than portfolio diversification or delegated monitoring. Consumers face a privately observed risk of becoming “impatient” and needing to consume early; because that risk is idiosyncratic and unverifiable, competitive trading in claims on a productive but illiquid technology cannot implement the allocation a social planner would choose, and decentralized markets merely reproduce autarky. A demand deposit contract — first-come, first-served, promising a fixed sum to any depositor who withdraws early — pools liquidity risk across depositors and can implement the planner’s optimal allocation as one Bayesian Nash equilibrium. The same contract, however, admits a second, strictly worse equilibrium in which every depositor withdraws early purely because each expects the others to, destroying investment in place and demonstrating that a bank run can be entirely self-fulfilling rather than information-driven. The paper further shows that suspension of convertibility restores uniqueness only when aggregate withdrawal demand is deterministic, and that government deposit insurance, financed by the state’s power to tax rather than by pre-funded reserves, eliminates the bad equilibrium even when withdrawal demand is stochastic.

1. Introduction

Diamond and Dybvig (1983) is a theoretical paper in the economics of financial intermediation. It belongs to the banking and financial-intermediation literature, and more precisely to the sub-field that asks why banking institutions exist at all — not as a matter of accounting convention, but as a response to a genuine economic friction. The paper is formally a general-equilibrium, mechanism-design exercise: it specifies an economy with a single technology and a single source of uncertainty, characterizes the allocation a planner would choose, and then asks which decentralized institutions can or cannot implement that allocation as an equilibrium of a non-cooperative game played by depositors.

The paper does not report an estimated coefficient, a t-statistic, or a sample of banks. It reports two propositions and builds the argument required to make them precise: first, that a particular contract — the demand deposit — can replicate an unattainable market outcome and, in doing so, explains why banks exist; second, that the very feature of the contract that makes it valuable (paying whoever shows up first, in full, until the money runs out) also makes it fragile, because it supports a second equilibrium in which the bank fails for no reason connected to its underlying assets. This is the paper's central intellectual move, and it is what makes it worth reading in 2026 rather than only citing: it is the founding formal statement of the idea that a bank run can be a rational, self-fulfilling coordination failure, not evidence of prior insolvency or of noisy information about asset quality.

Before the paper, banking theory did not lack informal accounts of runs — contagious panic, herd behavior, and the real damage of the Depression-era collapse of the U.S. banking system were well documented narratively, and Bryant (1980) had already sketched a closely related formal idea. What was missing was a rigorous general-equilibrium demonstration that (i) a bank contract is not an arbitrary institutional artifact but the solution to an actual risk-sharing problem, and (ii) the fragility of that contract is not a modeling defect to be assumed away, but a necessary consequence of the same first-come-first-served structure that makes the contract valuable in the first place. Diamond and Dybvig supply both halves in one integrated model, and then close the argument by showing which policy interventions do, and do not, resolve the fragility. That closing step — the comparison between suspension of convertibility and government deposit insurance — is what turns the paper from a description of a phenomenon into a normative statement about institutional design, and it is the reason the paper remains a direct reference point in the regulatory response to every banking panic since, including the March 2023 failures of Silicon Valley Bank and Signature Bank.

2. Research Problem and Hypothesis

2.1 Research Question

The paper asks two nested questions. The first is a pure risk-sharing question: when individual consumers face a privately observed, idiosyncratic risk of needing to consume earlier than expected, and the only productive technology available pays more the longer it is left in place, can any institution deliver more insurance against that risk than a competitive market in claims on the technology can deliver on its own? The second question is conditional on an affirmative answer to the first: if a particular contract — the demand deposit — can deliver that insurance, does it do so as the *only* possible outcome of depositors acting in their own interest, or does the same contract also support a second, collectively disastrous outcome supported purely by depositors' beliefs about each other's behavior? The paper's deepest claim is contained in the second question: multiplicity of equilibria is not a technical nuisance to be assumed away, but the direct and unavoidable counterpart of the contractual feature that makes the first-best allocation attainable at all.

2.2 Motivation

The question matters for two distinct reasons, one theoretical and one practical. Theoretically, the 1970s and early 1980s literature on financial intermediation (following Gurley and Shaw, 1960, and the portfolio-theoretic tradition) had explained banks chiefly as diversifiers or as delegated evaluators of borrower risk, but had not identified a distinct role for banks as providers of *liquidity insurance* to individually risky, privately informed consumers — a role that exists even when every loan in the bank's portfolio is riskless. Practically, the question of why banking panics recur, and whether they are symptoms of bad banks or a structural property of a useful institution, was — and remains — central to the design of banking regulation. A theory that locates the fragility in the contract itself, rather than in asset quality, implies a very different policy prescription (insure the contract) than a theory that locates it in insolvent or badly managed banks (liquidate the bank).

2.3 Existing Literature

At the time of writing, the closest formal antecedent was Bryant (1980), who had modeled banks as institutions offering deposit contracts to consumers facing idiosyncratic liquidity risk and noted the possibility of runs, but without deriving the risk-sharing optimum against which the deposit contract could be judged, and without separating a purely belief-driven run equilibrium from a run caused by adverse information about asset returns. Gurley and Shaw (1960) had established the broader view of financial intermediaries as transformers of asset characteristics (maturity, denomination, risk) between savers and borrowers, which supplies the institutional backdrop but not a formal account of why the specific liability structure of a bank — demandable, fixed-value deposits rather than tradable equity claims — is itself an equilibrium object. The empirical and narrative literature on the Great Depression banking panics (most influentially Friedman and Schwartz, 1963) had established that bank failures had real macroeconomic consequences, but

attributed the transmission mechanism primarily to the resulting contraction of the money supply, leaving open whether runs could also inflict damage through a channel unrelated to money — namely, the premature liquidation of productive investment.

2.4 Research Gap

What was missing, specifically, was a single model in which (a) the socially optimal allocation is derived from first principles as the solution to an explicit risk-sharing problem under private information, (b) a decentralized contract is shown to implement that optimum as one equilibrium, (c) the coexistence of a second, strictly inferior equilibrium is derived from the same primitives rather than assumed, and (d) the model is used to evaluate, rather than merely describe, competing policy responses — suspension of convertibility versus deposit insurance — on a common analytical footing.

2.5 Contribution

PAPER Diamond and Dybvig's stated contribution is exactly this integrated model: a formal demonstration that bank-like liability structures are the efficient response to uninsurable, privately observed liquidity risk, and a formal demonstration that the same structure is inherently exposed to a self-fulfilling run equilibrium whose likelihood and consequences can be altered by policy design. **CRITIQUE** The claimed contribution is substantially delivered within the boundaries the authors set for themselves — a single bank, a single riskless technology, no aggregate uncertainty in the baseline case — and the paper is explicit, rather than evasive, about which of those boundaries it does not relax. Whether the contribution extends cleanly beyond those boundaries (multiple banks, risky assets, aggregate shocks, information-based as opposed to purely sunspot-driven runs) is precisely the subject of the three decades of follow-on literature discussed in Section 9, and is not something the 1983 paper itself claims to resolve.

3. Theoretical Framework and Mechanism

3.1 Primitives: technology, types and preferences

The economy has three dates, $t = 0, 1, 2$, and a continuum of ex-ante identical consumers of unit mass, each endowed with one unit of the single good at date 0. There is one productive technology: one unit invested at date 0 returns one unit if liquidated at date 1, or $R > 1$ units if held to date 2. The technology is riskless — its return is not a random variable — which is a deliberate simplification the authors use to isolate liquidity risk from asset risk.

At date 1, each consumer privately learns her own type. With probability $t \in (0, 1)$ she is an *impatient* (type-1) consumer who values consumption only at date 1; with probability $1-t$ she is *patient* (type-2) and values the sum of consumption received at dates 1 and 2. By the law of large numbers over the continuum, t is also the realized fraction of impatient consumers in the baseline

version of the model, so there is no aggregate uncertainty about how many consumers will turn out to be impatient — a restriction the paper itself relaxes only partially, in the discussion of stochastic withdrawal demand (Section 3.5 below).

t	probability (and, absent aggregate uncertainty, realized fraction) of being an impatient, type-1 consumer.
R	gross return to the technology if held from date 0 to date 2, with $R > 1$.
c_1, c_2	consumption assigned to an impatient consumer at date 1, and to a patient consumer at date 2, respectively.
$u(\cdot)$	common, strictly increasing, strictly concave utility function, with $u'(0)=\infty$.
f	fraction of depositors already served when a given depositor reaches the bank window (the sequential service constraint).

3.2 Autarky, market trading, and why both are suboptimal

PAPER A consumer with no access to insurance simply stores the endowment for one period and consumes $c_1=1$ if impatient, or holds it for two periods and consumes $c_2=R$ if patient. Introducing a competitive market at date 1 in which consumers can trade claims to date-2 output does not improve on this outcome: because each consumer's type is private information, no contract can be written that is conditioned directly on type, and the only market-clearing price consistent with agents behaving optimally given their private information reproduces the autarky allocation exactly — type-1 consumers sell their date-2 claim for one unit of date-1 good, type-2 consumers buy such claims and get R . **EVIDENCE** The autarky/market point $(c_1, c_2)=(1, R)$ is not merely one candidate allocation but the specific point at which the planner's resource constraint below is satisfied with $c_1=1$, which is what makes the comparison between it and the planner's optimum exact rather than approximate.

CRITIQUE This is the paper's first and, in some ways, most important theoretical result, and it is easy to under-read: a complete market for a claim on the technology already exists in this economy, and it still fails to deliver insurance, because the underlying risk (impatience) is not a state that can be written into a contract — it is privately observed and unverifiable. The inefficiency here is therefore not the standard Arrow–Debreu missing-market story about an absent security; the security exists and trades. It is a private-information problem indistinguishable, formally, from the reason insurance markets more generally struggle to fully insure idiosyncratic, unverifiable risks.

3.3 The social planner's optimal risk-sharing allocation

A planner who could pool the entire endowment, observe realized types perfectly, and enforce any allocation would choose (c_1, c_2) to maximize the representative consumer's ex-ante expected utility, subject to feasibility. Because it is never optimal to give a patient consumer anything at date 1 (a unit consumed early is a unit not multiplied by R), the planner's problem reduces to two consumption levels:

$$\max_{c_1, c_2} t u(c_1) + (1-t) u(c_2) \quad \text{s.t.} \quad (1-t)c_2 = R(1 - t c_1) \quad (1)$$

The constraint says that whatever is not paid out to the t impatient consumers at date 1 stays invested, grows at rate R , and is split among the $1-t$ patient consumers at date 2. Forming the Lagrangian and differentiating with respect to c_1 and c_2 gives the planner's first-order condition:

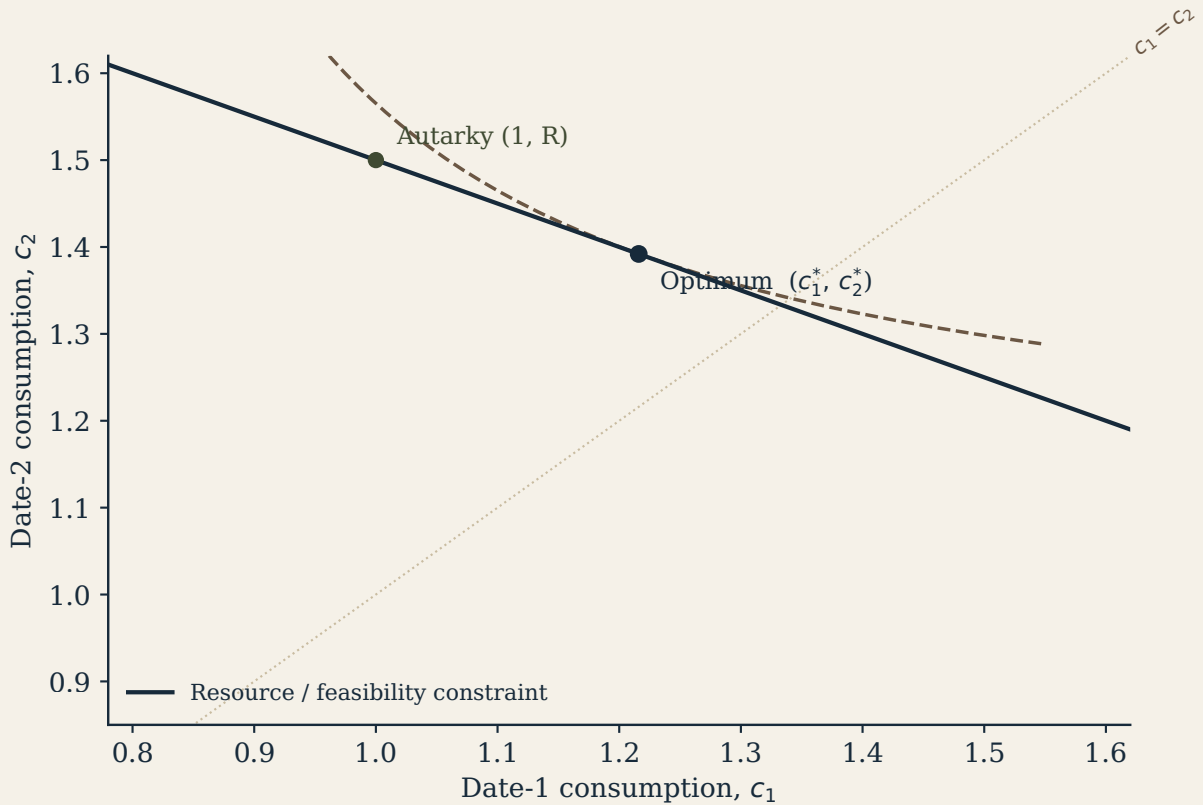
$$u'(c_1^*) = R \cdot u'(c_2^*) \quad (2)$$

which equates the marginal utility of an extra unit consumed early to the marginal utility of the R units it would have become had it instead been invested and consumed late. PAPER Diamond and Dybvig show that whenever relative risk aversion exceeds one everywhere — formally,

$$-cu''(c)/u'(c) > 1 \text{ for all } c \quad (3)$$

— the solution to (1)–(2) satisfies $1 < c_1^* < c_2^* < R$: the planner strictly raises early consumption above the autarky value of 1 and strictly lowers late consumption below the autarky value of R . Patient consumers accept a lower expected payoff than R in exchange for insurance against the possibility of turning out to be impatient; impatient consumers receive more than the one unit autarky would give them. This is risk-sharing exactly in the Arrow-Debreu sense, applied to a risk — one's own liquidity type — that autarky and market trading both leave uninsured.

Numerical illustration only — not reported in the paper. Diamond and Dybvig state the model and its comparative statics entirely in general functional form; the figure below solves a CRRA example ($u(c)=c^{1-\gamma}/(1-\gamma)$, $\gamma=3$, $t=0.25$, $R=1.5$) purely to give equations (1)–(2) a concrete, visually inspectable solution.

Figure 1. The planner's optimal allocation versus autarky, CRRA illustration

Solid line: the resource (feasibility) frontier, equation (1). Dashed line: an indifference curve of the representative consumer's ex-ante expected utility, tangent to the frontier at the optimum. Solving equations (1)–(2) numerically for $\gamma=3$, $t=0.25$, $R=1.5$ gives $c_1^* \approx 1.216$ and $c_2^* \approx 1.392$, against the autarky point $(1, 1.5)$. Source: author's own calculation, not from the original paper.

3.4 The demand deposit contract and the sequential service constraint

PAPER The planner's allocation cannot be handed out directly, because the planner cannot observe who is truly impatient. Diamond and Dybvig show that a specific decentralized contract — the demand deposit — can nonetheless implement (c_1^*, c_2^*) without ever verifying type, by exploiting depositors' own incentive to self-select. The bank commits, at date 0, to pay c_1^* to anyone who withdraws at date 1, in the order in which they arrive, until its assets are exhausted, and to divide whatever remains at date 2 equally among those who waited. This is the *sequential service constraint*: the payment a depositor receives can depend only on her position $f \in [0, 1]$ in the withdrawal queue, never on her true type or on the total number of people who will eventually withdraw, because neither is observable to the bank at the moment of payment. Formally, the payment schedule is

$$V_1(f) = c_1^* \quad (\text{pays every date-1 arrival, until reserves are gone}) \quad (4a)$$

$$V_2(f) = R(1 - fc_1^*) / (1 - f) \quad (4b)$$

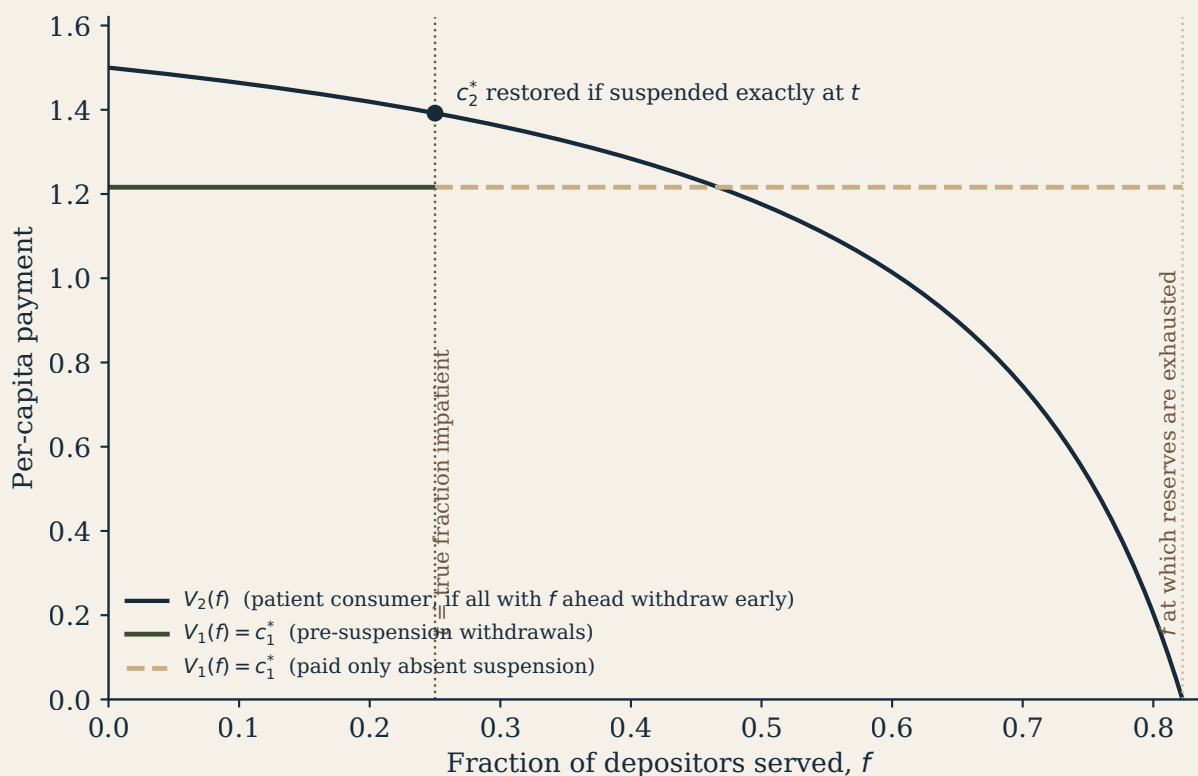
Equation (4b) is the per-capita payment to those who wait, given that a fraction f has already withdrawn c_1^* each. Since impatient consumers strictly prefer withdrawing early regardless of belief, and since $c_1^* < c_2^*$ makes waiting strictly better than withdrawing early for a patient consumer *provided she expects the fraction withdrawing early to equal exactly t* , the strategy profile “impatient consumers withdraw at date 1, patient consumers wait until date 2” is a Bayesian Nash equilibrium of the withdrawal game, and it delivers exactly the planner’s allocation (c_1^*, c_2^*) despite the bank never observing anyone’s type.

3.5 The bank-run equilibrium

PAPER The same contract admits a second equilibrium. If a patient consumer expects every other patient consumer to withdraw at date 1, then equation (4b) implies that the bank’s reserves are exhausted before date 2 — specifically, once a fraction $f = 1/c_1^*$ of the population has withdrawn c_1^* each, nothing remains for those still in the queue. A patient consumer who waits under this belief receives zero (or, in the version with partial liquidation of the technology before maturity, something below c_1^*), which is worse than joining the queue and receiving c_1^* with some probability of being served before reserves run out. Running is therefore each patient consumer’s best response to the belief that every other patient consumer will run — a second, self-fulfilling Bayesian Nash equilibrium of exactly the same contract, the same technology, and the same fundamentals as the good equilibrium in Section 3.4. **CRITIQUE** This is the paper’s defining theoretical achievement: the run is not caused by a shock to R , to the technology, or to anyone’s type distribution — it is caused purely by a shift in beliefs, and nothing internal to the model selects one equilibrium over the other. What triggers a shift in beliefs (a “sunspot”) is left entirely outside the model, which is both the source of the paper’s theoretical power and, as Section 8 discusses, its principal empirical limitation.

3.6 Suspension of convertibility

PAPER If the fraction of truly impatient consumers, t , is known with certainty in advance, the bank can eliminate the run equilibrium entirely by committing to suspend payments once a fraction t of depositors has withdrawn, refusing to serve anyone beyond that point at date 1 regardless of how large the queue becomes. Because the queue is now capped at exactly the number of genuinely impatient consumers, a patient consumer can no longer be deprived of c_2^* by a run — trying to withdraw early yields nothing once the cap binds, while waiting is guaranteed to yield c_2^* . Waiting becomes a dominant strategy for patient consumers, and the run equilibrium is destroyed, leaving the good equilibrium as the unique outcome.

Figure 2. Suspension of convertibility: the payment schedule as a function of queue position

Same CRRA calibration as Figure 1. $V_2(f)$ (navy) is the per-capita payoff to a patient consumer who waits, given that a fraction f has already withdrawn c_1^* each — it collapses toward zero as f approaches the point at which reserves are exhausted ($f \approx 0.82$ in this calibration). Suspending payment exactly at $f=t$ caps the queue before this decline can occur and restores c_2^* with certainty. Source: author's own calculation, not from the original paper.

PAPER The authors are explicit that this remedy depends on the planner (the bank) knowing t in advance. If the fraction of consumers who turn out to be impatient is itself random — if there is aggregate, not merely idiosyncratic, uncertainty about liquidity demand — a suspension threshold fixed in advance either turns away genuinely impatient consumers (if realized demand exceeds the threshold) or fails to prevent a run (if it is set high enough to accommodate the worst case, but a run can still exhaust reserves before that cap binds). Suspension of convertibility is therefore, in the paper's own analysis, a solution calibrated to a special case rather than a general one.

3.7 Government deposit insurance

PAPER The paper's second and, in its framing, more robust remedy is government-provided deposit insurance, financed not by pre-funded reserves but by the government's power to levy taxes ex post. The government guarantees every date-1 withdrawer c_1^* , adjusting the tax levied on withdrawals (or, symmetrically, rebating unused capacity) once the realized fraction of withdrawals is observed, so that the bank's ability to pay c_2^* to those who wait is never put at risk by how many depositors happen to withdraw early — including a fraction driven purely by panic. Because the payoff to waiting is now invariant to what a patient consumer believes other depositors will do, running stops being a best response to any belief, and the insured contract implements the planner's allocation as the *unique* equilibrium, even when t is stochastic.

The paper is careful to explain why a privately supplied insurance contract cannot achieve the same result on its own: a private insurer, lacking the power to tax, must back its guarantee with pre-committed reserves large enough to cover the worst realization of withdrawal demand, which is exactly the illiquid-reserve problem the deposit contract was designed to avoid in the first place. The government's taxing power substitutes for reserves, and — crucially for the policy conclusion — the insurance is essentially costless if a run never actually occurs, since no reserves need to be diverted from the productive technology in the good state of the world.

4. Empirical Content and Testability

PAPER Diamond and Dybvig (1983) contains no data, no estimation, and no numerical example of its own — it is a pure theory paper, and the paper's propositions are established by construction and comparative statics rather than by measurement against observed banking outcomes. Section 29 of the governing review framework for this library calls for a Data and Empirical Strategy section in every review; for a paper of this type that section is necessarily adapted rather than force-fitted, and is used here to state plainly what the model does and does not predict that can, in principle, be taken to data.

4.1 What the model is silent on

The model does not specify a probability that a sunspot shift in beliefs occurs, does not identify observable pre-conditions that make a run more or less likely, and does not distinguish, within its own structure, between a run driven by pure panic and a run driven by adverse information about the bank's true solvency — a distinction the paper deliberately sets aside by making the technology riskless. As a direct consequence, the 1983 model in isolation is not, by itself, testable against a cross-section of historical bank failures: it predicts that runs *can* occur without predicting *when*.

4.2 How the literature has since supplied testable content

EXTENSION Later work, discussed fully in Section 9, supplies exactly this missing content by embedding the Diamond–Dybvig coordination structure inside models with observable fundamentals (Goldstein and Pauzner, 2005) or by testing, directly on historical or contemporary deposit-flow data, whether withdrawal behavior around a bank's distress episode looks more consistent with a panic-based run (withdrawals uncorrelated with individually observable signals of bank health) or an information-based run (withdrawals concentrated among depositors with better information, or following public news of asset losses). Gorton (1988) and Calomiris and Gorton (1991) find that historical U.S. banking panics were statistically associated with business-cycle indicators consistent with an information-based, rather than purely sunspot-driven, component, which is the paper's cleanest confrontation with the historical record even though the confrontation was carried out by others, using a data set the 1983 paper never proposed to construct itself.

5. Results

5.1 Proposition 1 — markets alone cannot deliver the optimum

PAPER What was tested: whether a competitive market for date-2 claims, open to consumers who have already privately learned their type, converges to the risk-sharing optimum characterized by equations (1)–(2). **PAPER** What was found: it does not — the market allocation coincides exactly with autarky, $(c_1, c_2) = (1, R)$, because private information about type prevents any type-contingent contract from being written and enforced through anonymous trading. **CRITIQUE** The magnitude of the gap between autarky and the optimum is exactly the risk-sharing gain foregone: in the CRRA illustration of Section 3.3, patient consumers give up $R - c_2^* \approx 0.108$ units of expected date-2 consumption (about 7 percent of the autarky value) in exchange for impatient consumers gaining $c_1^* - 1 \approx 0.216$ units (a 22 percent increase) — economically large numbers for what is, ex ante, a symmetric insurance arrangement among identical agents. This result supports the paper’s hypothesis directly and is not conditional on any auxiliary assumption beyond private information and risk aversion exceeding one.

5.2 Proposition 2 — the demand deposit contract implements the optimum as an equilibrium

PAPER What was tested: whether a specific, simply described contract (first-come-first-served, fixed early payment) can implement (c_1^*, c_2^*) without the bank ever observing type. **PAPER** What was found: yes, as one Bayesian Nash equilibrium of the induced withdrawal game, exploiting only the ordering $c_1^* < c_2^*$ established in Section 3.3. **CRITIQUE** This is the paper’s positive, constructive result and it is fully general within the model’s assumptions — it does not depend on the specific CRRA calibration used for illustration in this review, only on the sign of the risk-aversion condition in equation (3). It is, however, an existence result, not a uniqueness result, which is precisely the subject of the next finding.

5.3 Proposition 3 — the same contract also supports a run equilibrium

PAPER What was tested: whether “all patient consumers withdraw early” is also consistent with individually rational behavior under the same contract. **PAPER** What was found: yes — it is a second Bayesian Nash equilibrium, strictly Pareto-dominated by the first, in which patient consumers receive less than c_1^* in expectation and the technology is prematurely liquidated. **CRITIQUE** Economically, this result is the paper’s most consequential: it says that the mere existence of a fragile institution is not evidence of mismanagement, because fragility of exactly this kind is unavoidable in *any* contract that delivers the risk-sharing gains of Proposition 2 through a sequential-service, fixed-early-payment structure. This is a strong and, on reflection, quite general claim, and Section 8 interrogates how far it can be pushed.

5.4 Proposition 4 — suspension works only without aggregate uncertainty; insurance works more generally

PAPER What was tested: two candidate remedies — a fixed suspension threshold and government-financed deposit insurance — against two informational environments, one with a known t and one with stochastic aggregate withdrawal demand. **PAPER** What was found: suspension restores uniqueness (eliminates the run equilibrium) only in the first environment; deposit insurance restores uniqueness in both, because it does not rely on the government knowing t in advance — only on being able to observe and tax realized withdrawals ex post. **CRITIQUE** The economic significance of this result lies less in the individual propositions than in their juxtaposition: the paper does not simply recommend deposit insurance as a good idea, it derives the specific informational condition (uncertainty about aggregate liquidity demand) under which suspension, the older and at the time more commonly used tool, provably fails and insurance provably does not. That is a sharper and more falsifiable policy claim than either remedy considered on its own.

6. Robustness of the Mechanism

The paper contains no empirical robustness checks in the statistical sense — there is no alternative sample, no alternative estimator, no placebo test, because there is no estimation. What the paper does instead, and what this section evaluates in the same concern-test-result structure used elsewhere in this library for empirical papers, is vary its own assumptions internally and show which results survive.

Source: author's synthesis of arguments made across Sections II–IV of Diamond and Dybvig (1983); not a reproduction of any table in the original paper, which contains none.

Table 1. Internal robustness of the paper's core claims

Concern	Internal test	Result	Resolves the concern?
Does the run equilibrium depend on risk aversion exceeding one?	The multiplicity argument (Section 3.5) is re-derived using only $c1^* < c2^*$, not the specific inequality in equation (3).	Multiplicity survives; only the size of the risk-sharing gain (Section 5.1) depends on the risk-aversion condition.	Yes
Does suspension require exact knowledge of t ?	The authors replace deterministic t with a stochastic aggregate withdrawal fraction and re-examine whether a fixed threshold still blocks all profitable running.	It does not; a threshold sufficient to accommodate the worst realization of impatience no longer prevents a run at lower realizations.	Partially — confirms the limitation rather than resolving it
Is deposit insurance robust to the same stochastic t ?	The government's tax-and-rebate rule is defined as a function of the realized, not anticipated, withdrawal fraction.	Uniqueness of the good equilibrium is preserved for any realization of t .	Yes

Concern	Internal test	Result	Resolves the concern?
Can a private insurer replicate the government's guarantee?	The authors substitute a reserve-constrained private insurer for the government in the same contracting environment.	A private insurer must pre-fund reserves against the worst case, reintroducing the illiquidity problem the deposit contract was built to avoid.	No — confirms the asymmetry favors public insurance

CRITIQUE What this internal robustness exercise cannot do — because the paper does not attempt it — is establish how *likely* a shift to the run equilibrium is in practice, whether real-world suspension clauses (which were common in pre-Federal-Reserve U.S. banking and remain common in money-market-fund gating rules) perform as the model predicts, or whether deposit insurance’s theoretical robustness to aggregate uncertainty survives the introduction of the one channel the paper deliberately excludes: moral hazard in the bank’s own portfolio choice once its liabilities are insured. That last omission is arguably the paper’s single most consequential unmodeled channel and is returned to in Sections 7 and 10.

7. Critical Evaluation

7.1 What the paper actually establishes

ESTABLISHED

That, under private information about idiosyncratic liquidity risk, a first-come-first-served demand deposit contract can implement the socially optimal risk-sharing allocation as one equilibrium, and that the same contract necessarily also supports a strictly worse, purely belief-driven run equilibrium. Both halves follow directly from the stated assumptions and are proved, not merely illustrated.

SUPPORTED BUT CONDITIONAL

That government deposit insurance dominates suspension of convertibility as a policy response. This holds conditional on the government’s taxing power being genuinely more credible than a private insurer’s reserve commitment, and conditional on the moral-hazard costs of insuring bank liabilities being small relative to the run-prevention benefit — a comparison the paper does not itself carry out, because it does not model risky bank assets or portfolio choice at all.

SUGGESTIVE

That bank runs in general — as opposed to the specific run equilibrium of this model — are “real” phenomena with no informational content and no connection to fundamentals. The model is consistent with this view because it is built to isolate a purely sunspot-driven mechanism, but it does not establish that historical runs were in fact of this kind rather than information-driven; it shows only that a purely belief-driven run is theoretically possible and, within the model, payoff-relevant.

NOT ESTABLISHED

Anything about which banks are more or less likely to be run on, how large deposit insurance premia should be, how the results change with risky bank assets, multiple banks, or an interbank market, or how large the moral-hazard cost of insurance is relative to the run-prevention benefit it buys. None of these questions are within the model’s scope, and the paper does not claim otherwise.

7.2 Strengths

PAPER The authors themselves emphasize, correctly, that the model isolates liquidity-risk-based banking from asset-risk-based banking by making the technology riskless — a deliberate simplification that lets them attribute fragility entirely to contract structure rather than to portfolio quality. **CRITIQUE** Independently of that stated strength, three further features of the paper stand out. First, its parsimony: the entire argument rests on a three-date, two-type economy with one technology, yet it generates both a positive theory of bank existence and a positive theory of bank fragility from the same primitives — very little is assumed that is not used. Second, its treatment of multiplicity as the object of study rather than an inconvenience: many contemporaneous models of coordination failure simply posited multiple equilibria; this paper derives multiplicity from the contract designed to solve the risk-sharing problem, which is a stronger and more informative result. Third, the clarity with which the paper separates two policy tools (suspension, insurance) that were, at the time, often discussed together, and derives the exact informational condition that breaks the equivalence between them.

7.3 Weaknesses and independent critique

The paper’s own limitations section is unusually candid for a theory paper of its era, and this review separates what the authors flag themselves from weaknesses identified independently below.

PAPER The authors note that the riskless technology, while useful for isolating the liquidity-insurance channel, abstracts entirely from the bank’s choice of loan portfolio risk, and that a single-bank framework leaves no role for interbank liquidity sharing, which a multi-bank extension would introduce.

CRITIQUE Beyond what the authors flag, five weaknesses merit independent emphasis, in the problem→why it matters→potential bias→testability structure this library applies consistently.

Moral hazard is entirely absent. The paper's central policy conclusion — that government deposit insurance dominates suspension — is derived in a world where insuring the bank's liabilities has no effect on the bank's asset-side behavior. This matters because the single strongest real-world objection to blanket deposit insurance is precisely that it changes incentives on the asset side (Kareken and Wallace, 1978, make this point contemporaneously). The bias runs in one direction: the paper's comparison overstates the net benefit of insurance relative to suspension, because it credits insurance with eliminating a cost (run risk) without debiting it for a cost the model cannot see (excess risk-taking). This is testable in principle by comparing risk-taking and run incidence across deposit-insurance regimes, which later empirical work (Demirgüç-Kunt and Detragiache, 2002) undertakes directly.

Equilibrium selection is left entirely exogenous. Nothing inside the model assigns a probability to the run equilibrium being played; it is either assumed away (the good equilibrium) or assumed to occur (the bad one), with no mechanism connecting the two to observable circumstances. This matters because a theory that cannot, even in principle, say when a run is more or less likely is of limited direct use for calibrating regulation, and the gap was not closed until the global-games literature (Section 9) supplied an equilibrium-selection device two decades later.

The sequential service constraint is assumed, not derived. The entire fragility result depends on payments being conditioned only on queue position; if the bank could instead condition payments on some other observable signal, or if depositors could costlessly renegotiate after arriving, the run equilibrium could in principle be engineered away. Wallace (1988) shows this assumption is doing real work and is not an innocuous modeling convenience.

The model has no aggregate risk in its baseline case, yet the deposit-insurance result is strongest precisely when aggregate risk is present. This is not a contradiction, but it does mean the paper's most policy-relevant comparison (Section 5.4) is carried by an extension sketched more briefly than the baseline model that occupies most of the paper's length.

The paper is silent on why demandable debt, rather than some other security, is the contract banks actually use. It shows that a first-come-first-served fixed-payment contract can implement the optimum; it does not show that no other security (for example, a tradable equity claim with a market-determined price) could do so with less fragility. This is exactly the question Jacklin (1987) takes up, with an answer that qualifies the paper's implicit claim that demand deposits are the *only* efficient response to the underlying risk-sharing problem.

8. Contradictions and Edge Cases

INFERENCE What happens as $R \rightarrow I$? As the return to holding the technology to maturity converges to the return from early liquidation, the gains from trade underlying the entire model vanish: the resource constraint in equation (1) collapses toward $c_2 = 1 - tc_1 / (1 - t)$ with slope approaching $-t / (1 - t)$ rather than $-Rt / (1 - t)$, and the optimal contract converges to autarky, $c_1^* \rightarrow 1$. In this limit the demand deposit contract still exists formally, but it insures against a risk that no longer carries an economic cost, and the run equilibrium, while still technically present, destroys a

correspondingly small surplus. This is a useful sanity check: the model does not manufacture fragility out of nothing — fragility is the price of a risk-sharing gain that must itself be positive for the mechanism to matter.

INFERENCE **What happens as risk aversion falls below the threshold in equation (3)?** If relative risk aversion is everywhere less than one, the ordering $c_1^* < c_2^*$ that makes waiting attractive for patient consumers under the good equilibrium can fail to hold strictly, undermining the incentive structure that supports Proposition 2. In the boundary log-utility case (relative risk aversion equal to one), the risk-sharing gain from the contract shrinks toward the autarky allocation from another direction than the $R \rightarrow I$ case above, illustrating that the model's central result is not a free lunch: it requires a genuine preference for consumption smoothing across dates, not merely an unequal endowment.

INFERENCE **What happens at the extremes of t ?** As $t \rightarrow 0$, almost no one is impatient, the risk-sharing problem becomes trivial, and the resource constraint approaches the vertical line $c_2 \approx R$; the model still technically supports a run equilibrium, but it would require virtually the entire patient population to coordinate on running to exhaust a bank holding almost fully invested assets, making the bad equilibrium correspondingly harder to sustain as a plausible focal point even though it remains, formally, an equilibrium. As $t \rightarrow 1$, almost everyone is impatient, the planner's allocation converges toward paying everyone close to 1 at date 1, and the distinction between the two equilibria becomes economically small because there is little late consumption left to expropriate. The model's fragility result is therefore most economically consequential at intermediate t , which is also the empirically relevant range for retail banking.

CRITIQUE **Could another mechanism generate the same observed pattern (a bank failing suddenly, with a queue of withdrawing depositors)?** Yes, and this is the model's central identification problem when it is taken outside pure theory. An information-based run — in which depositors observe or infer a negative signal about the bank's true (risky) asset quality and withdraw rationally on that basis — produces an observationally similar queue of withdrawals without requiring any coordination failure at all. Because the 1983 model assumes a riskless technology by construction, it cannot itself distinguish a sunspot-driven run from an information-driven one; that distinction has to be imported from a different (and later) class of models, most influentially Chari and Jagannathan (1988) and the global-games literature discussed next.

CRITIQUE **Does the multiplicity result survive if depositors can trade their deposit claims on a secondary market, rather than only withdraw in a physical queue?** This is precisely the question Jacklin (1987) poses. If deposit claims are tradable and the associated equilibrium price can move to reflect aggregate information, some of the coordination-failure logic that drives Diamond and Dybvig's multiplicity weakens, because a patient consumer with a tradable claim need not join a physical queue to avoid the worst outcome — she can sell. Whether this fully eliminates the run equilibrium, or merely changes its form, depends on assumptions about the correlation between aggregate and idiosyncratic risk that lie outside the 1983 paper's own scope.

9. Further Literature

Table 2. Placing the paper in the literature

Paper	Shared idea	Relationship
Bryant (1980)	Deposit contracts as a response to idiosyncratic liquidity risk; possibility of runs	Close formal predecessor; Diamond and Dybvig supply the risk-sharing optimum and the multiple-equilibria proof that Bryant's sketch lacked
Gurley and Shaw (1960)	Financial intermediaries as transformers of asset characteristics	Broader institutional foundation the paper builds a specific liquidity-insurance mechanism onto
Kareken and Wallace (1978)	Deposit insurance and bank portfolio risk	Contemporaneous critique of unconditional deposit insurance on moral-hazard grounds — the channel Diamond and Dybvig's riskless-technology assumption sets aside
Jacklin (1987)	Demand deposits versus equity claims as liquidity-insurance devices	Directly qualifies the 1983 result by showing tradable equity can sometimes dominate demand deposits, depending on the correlation of risks
Postlewaite and Vives (1987)	Endogenizing the probability that a run occurs	Extends the model to give the previously exogenous run probability a determinate, non-zero value
Wallace (1988)	The role of the sequential service constraint	Shows the constraint is essential to the fragility result, not an incidental modeling choice
Chari and Jagannathan (1988)	Information-based, rather than sunspot-based, bank runs	Competing mechanism generating observationally similar runs from a different, information-driven cause
Calomiris and Kahn (1991)	Demandable debt as a monitoring device	Alternative rationale for demand deposits (disciplining bank managers) rather than pure liquidity insurance
Gorton (1988); Calomiris and Gorton (1991)	Empirical incidence of historical U.S. banking panics	Confronts the model's predictions with the historical panic record, finding evidence consistent with an information-based component
Green and Lin (2003)	Mechanism-design refinement of the original model	Shows uniqueness can sometimes be restored even without insurance under richer information structures, qualifying the original multiplicity result
Peck and Shell (2003)	Runs under correlated types and the sequential service constraint	Extends multiplicity results to a setting closer to the original paper's own assumptions than Green and Lin's
Goldstein and Pauzner (2005)	Global-games equilibrium selection applied to bank runs	Supplies the equilibrium-selection mechanism the 1983 model lacks, making run probability a function of observable fundamentals

Paper	Shared idea	Relationship
Ennis and Keister (2009)	Time-consistency of suspension policies	Shows a government or bank may lack the ability to credibly commit to suspension once a run is under way, an implementation problem the 1983 paper does not address
Allen and Gale (1998, 2000)	Contagion across interconnected banks	Extends the single-bank framework to a banking system, reintroducing the interbank channel the 1983 paper explicitly leaves out
Diamond and Rajan (2001)	Bank capital structure and fragility	Endogenizes the demandable-debt/equity mix the 1983 paper takes as given

10. Research Gaps

10.1 Paper-level gaps

The 1983 model leaves the probability of a run exogenous; it does not model bank asset risk, multiple banks, or an interbank market; and it does not endogenize the bank’s capital structure — why deposits rather than some blend of debt and equity finance the technology.

10.2 Literature-level gaps

Three decades on, the literature has not fully settled the relative empirical importance of sunspot-driven versus information-driven runs, nor reached consensus on how much of the moral-hazard cost of deposit insurance is offset by its run-prevention benefit across different regulatory regimes and levels of supervisory quality.

10.3 Methodological gaps

Better identification of run probability as a function of bank-specific, ex-ante observable fundamentals (rather than as a free parameter) remains an active area; better high-frequency, depositor-level data on withdrawal timing — increasingly available in digital-banking and blockchain settings — could discriminate between competing mechanisms far more directly than the aggregate, low-frequency data available to the 1980s empirical banking-panic literature.

11. Research Opportunities

Idea 1 — Disentangling panic-driven from information-driven runs using on-chain stablecoin and DeFi lending-pool data

RESEARCH QUESTION: When over-collateralized DeFi lending pools or asset-backed stablecoins experience sudden mass redemptions, is withdrawal timing better explained by a Diamond–Dybvig-style coordination failure or by information-based withdrawal ahead of an observable collateral shortfall?

MOTIVATION: Blockchain settings provide something no historical banking panic offers: a fully observable, timestamped, depositor-level record of every withdrawal alongside the real-time collateral ratio, allowing the two competing mechanisms to be separated cleanly.

HYPOTHESIS: Withdrawal intensity accelerates discontinuously around observable collateral-ratio thresholds under information-based runs, but shows no such discontinuity — only clustering in time — under a pure coordination failure.

ECONOMIC MECHANISM: The sequential-service, first-come-first-served redemption structure of many DeFi lending pools mirrors the 1983 model’s payment schedule (equations 4a–4b) closely enough to make the model directly applicable rather than merely analogous.

LITERATURE CONNECTION: Directly tests the Diamond and Dybvig (1983) versus Chari and Jagannathan (1988) distinction with data neither could have used.

DATA: Public blockchain transaction logs for major lending protocols and asset-backed stablecoins around historical depeg or bank-run episodes (2022–2023).

VARIABLES: Withdrawal timestamps and amounts, real-time collateralization ratio, on-chain wallet-level sophistication proxies, protocol governance announcements.

METHOD: Event-study and discontinuity design around collateral-ratio thresholds, combined with a structural estimation of the Diamond–Dybvig run-threshold model.

IDENTIFICATION STRATEGY: Exogenous, protocol-determined collateral-ratio thresholds provide a sharp discontinuity design unavailable in traditional banking data.

EXPECTED CONTRIBUTION: The first clean, individual-level empirical separation of the two run mechanisms the theoretical literature has debated since 1988.

MAIN RISK: DeFi lending pools differ structurally from insured retail deposits (no deposit insurance, algorithmic rather than discretionary suspension), which may limit external validity to traditional banking.

Idea 2 — A global-games fragility index calibrated to bank balance sheets

RESEARCH QUESTION: Can the equilibrium-selection mechanism of Goldstein and Pauzner (2005) be operationalized into a bank-level fragility index using only publicly observable balance-sheet data, and does it predict which banks actually experienced runs (e.g., in March 2023)?

MOTIVATION: The original model cannot say which banks are more run-prone; the global-games refinement can, in principle, but has rarely been calibrated to real balance sheets at scale.

HYPOTHESIS: Banks with a higher ratio of uninsured deposits to liquid assets, and a noisier public signal of asset quality, have a higher global-games-implied run probability, and this index predicts realized 2023 deposit outflows better than simple liquidity ratios alone.

ECONOMIC MECHANISM: Combines the 1983 payoff structure (equations 4a–4b) with a noisy public signal of fundamentals to generate a unique, threshold-based equilibrium rather than the original model's two extremes.

LITERATURE CONNECTION: Bridges Diamond and Dybvig (1983) and Goldstein and Pauzner (2005) with the applied bank-fragility literature.

DATA: U.S. FDIC call-report balance-sheet data; uninsured deposit shares; 2023 deposit-outflow data for regional banks.

VARIABLES: Uninsured deposit share, unrealized securities losses, deposit concentration, public signal noise proxies (analyst coverage, credit-default-swap spreads).

METHOD: Structural calibration of the global-games threshold model, followed by an out-of-sample predictive test against realized 2023 outflows.

IDENTIFICATION STRATEGY: Cross-sectional variation in balance-sheet composition across regional banks with otherwise similar business models.

EXPECTED CONTRIBUTION: A forward-looking, theory-disciplined fragility measure usable by supervisors, rather than a purely backward-looking capital ratio.

MAIN RISK: Global-games predictions are sensitive to the assumed precision of the public signal, which is not directly observable and would need to be estimated or proxied.

Idea 3 — Redemption gates in open-end bond funds as a test of suspension of convertibility

RESEARCH QUESTION: Do redemption gates and swing pricing in open-end mutual funds reduce run-like outflows the way the 1983 model predicts suspension of convertibility should, or do they instead accelerate pre-emptive withdrawal, as Ennis and Keister's (2009) commitment critique would suggest?

MOTIVATION: Post-2008 concerns about liquidity mismatch in bond funds led regulators to require exactly the kind of suspension tool the 1983 paper analyzes, in a setting with excellent daily flow data.

HYPOTHESIS: Funds with credible, rules-based gating provisions experience smaller pre-emptive outflows ahead of stress episodes than funds with discretionary gating, consistent with the model's emphasis on commitment.

ECONOMIC MECHANISM: A rules-based gate plays the role of the 1983 model's known, credible suspension threshold; a discretionary gate resembles the stochastic- t case in which suspension is shown to be ineffective.

LITERATURE CONNECTION: Tests the suspension side of Diamond and Dybvig (1983) together with Ennis and Keister's (2009) commitment critique.

DATA: Daily fund-flow data for open-end bond mutual funds around 2020 (COVID) and 2022 (rate-shock) stress episodes, by gating-provision type.

VARIABLES: Daily net flows, gate/swing-pricing provisions, fund liquidity composition, stress-period indicators.

METHOD: Difference-in-differences comparing rules-based versus discretionary gating funds around stress episodes.

IDENTIFICATION STRATEGY: Cross-sectional variation in gating-provision design that is fixed prior to the stress episode, limiting reverse causality.

EXPECTED CONTRIBUTION: A direct empirical test of the suspension mechanism in a modern, data-rich setting the original 1983 authors could not have observed.

MAIN RISK: Fund investors differ from retail depositors in sophistication and outside options, which may weaken the analogy to the original model's consumer base.

Idea 4 — Endogenizing bank capital structure alongside the run equilibrium

RESEARCH QUESTION: How does the optimal mix of demandable deposits and loss-absorbing equity change once the bank's capital structure is chosen to trade off the Section 3 risk-sharing gain against run risk and regulatory capital requirements?

MOTIVATION: The 1983 paper takes the deposit contract as given; post-Basel III capital regulation makes the capital-structure margin first-order for actual bank behavior.

HYPOTHESIS: The equity share that minimizes total social cost rises with the volatility of aggregate withdrawal demand and falls with the credibility of deposit insurance, tracing out a testable relationship between insurance generosity and observed bank leverage.

ECONOMIC MECHANISM: Extends equations (1)–(2) to a three-claim economy (deposits, equity, illiquid assets) building on Diamond and Rajan (2001).

LITERATURE CONNECTION: Bridges Diamond and Dybvig (1983) and Diamond and Rajan (2001) with the bank-capital-regulation literature.

DATA: Cross-country bank capital ratios and deposit-insurance generosity indices (e.g., World Bank deposit insurance database).

VARIABLES: Equity-to-assets ratio, deposit insurance coverage limit and funding structure, measures of withdrawal-demand volatility.

METHOD: Structural model calibrated separately by country/regime, tested against a cross-country panel regression of leverage on insurance generosity.

IDENTIFICATION STRATEGY: Cross-country and cross-time variation in deposit insurance design (coverage limits, funding mechanism) as a source of plausibly exogenous variation in insurance generosity.

EXPECTED CONTRIBUTION: A unified, testable link between deposit-insurance design and observed bank capital structure, filling the gap identified in Section 10.1.

MAIN RISK: Bank capital structure is driven by many regulatory and tax factors beyond run risk, complicating clean identification of the mechanism specific to this model.

Idea 5 — A cross-country natural-experiment test of the moral-hazard offset to deposit insurance

RESEARCH QUESTION: Across countries and over time, does the introduction or expansion of deposit insurance reduce measured run/panic incidence by more or less than it increases measured bank risk-taking, and what is the net effect on financial stability?

MOTIVATION: This is the direct empirical resolution of the tension flagged in Section 7.3 between the paper’s own conclusion and the moral-hazard channel it excludes by assumption.

HYPOTHESIS: Deposit insurance reduces run-related bank failures but raises non-run-related failures via increased risk-taking, with the net effect on total failure rates depending on supervisory quality — following, and updating with post-2010 data, Demirgüç-Kunt and Detragiache (2002).

ECONOMIC MECHANISM: Combines the 1983 model’s run-elimination result with an unmodeled asset-risk channel, made explicit via a simple moral-hazard extension.

LITERATURE CONNECTION: Directly extends Kareken and Wallace (1978) and Demirgüç-Kunt and Detragiache (2002) with a longer, more recent panel.

DATA: Cross-country panel of deposit insurance adoption dates and design features, bank failure rates, and supervisory-quality indices (World Bank, IMF Financial Soundness Indicators).

VARIABLES: Deposit insurance adoption/coverage, bank failure rates decomposed by cause where possible, supervisory quality, bank-level risk-weighted asset growth.

METHOD: Staggered difference-in-differences around deposit-insurance adoption or coverage-limit changes, interacted with supervisory quality.

IDENTIFICATION STRATEGY: Staggered adoption timing across countries as quasi-natural experiments, with supervisory quality as a moderating variable.

EXPECTED CONTRIBUTION: An updated, direct quantification of the net welfare effect of deposit insurance, addressing the single largest unmodeled channel in the original paper.

MAIN RISK: Deposit insurance adoption is rarely random and often co-occurs with other regulatory reforms, threatening the exclusion restriction needed for causal identification.

11.1 Ranking

EXTENSION Ranked by novelty, feasibility, data availability, methodological quality, and economic importance, the strongest three ideas are, in order: **Idea 1** (DeFi/stablecoin redemption data), because it is the only proposal offering a genuinely new, fully observable data source capable of cleanly separating two mechanisms the literature has debated on indirect evidence for nearly forty years; **Idea 5** (cross-country moral-hazard offset), because it directly confronts the paper’s single largest unmodeled channel with the most mature and best-populated data set among the five proposals; and **Idea 2** (global-games fragility index), because it converts an influential but rarely operationalized theoretical refinement into a measure directly usable against the 2023 regional-bank episode, with a clear, near-term out-of-sample test. Ideas 3 and 4 are worthwhile but face weaker identification (fund investors are not close analogues of retail depositors; capital structure has many non-run-related determinants) than the top three.

12. Learning Framework

12.1 Prerequisite Concepts

Expected-utility theory and constant relative risk aversion; the Arrow–Debreu notion of a complete market and what private information about type breaks in that framework; Bayesian Nash equilibrium and the concept of multiple equilibria supported by self-fulfilling beliefs; the basic mechanics of financial intermediation as maturity transformation.

12.2 Core Concepts Learned

That a bank contract can be understood as the solution to a risk-sharing mechanism-design problem under private information, rather than as an arbitrary institutional form; that fragility and risk-sharing value can be two faces of the same contractual feature rather than a design flaw to be engineered away; and that a coordination failure among individually rational agents can be entirely self-fulfilling, with no informational or fundamental trigger required.

12.3 Important Equations

Equation (2), $u'(c_1^*) = Ru'(c_2^*)$, is the single most important relationship in the paper: it is the standard consumption-smoothing Euler-type condition, applied here to characterize optimal insurance rather than optimal saving. Equations (4a)–(4b), the sequential-service payment schedule, are the mechanical heart of the multiplicity result and worth being able to reproduce and manipulate directly, since nearly the entire subsequent literature (Sections 9 and 11) modifies exactly this schedule.

12.4 Methods Learned

Solving a planner’s problem subject to a resource constraint to characterize a risk-sharing optimum; constructing a decentralized mechanism and verifying it implements that optimum as a Bayesian Nash equilibrium; and, separately, demonstrating that a second, undesirable equilibrium exists under the same primitives — a proof technique (construct-and-verify multiplicity) used throughout the coordination-failure literature in macroeconomics and finance.

12.5 Questions to Test Understanding

Comprehension. Can you state, without notes, why a competitive market for date-2 claims fails to deliver the planner’s allocation even though such a market exists and clears? Can you derive equation (2) from the planner’s problem in equation (1)?

Analysis. Can you explain precisely why the sequential service constraint, rather than the size of R or t , is what makes the run equilibrium possible? Can you explain why suspension of convertibility fails once t is stochastic, using equation (4b) directly?

Research. Can you design an empirical test that would distinguish a Diamond–Dybvig sunspot-driven run from a Chari–Jagannathan information-driven run using data that did not exist in 1988? Can you extend the model to a two-bank economy with an interbank market and characterize whether the multiplicity result survives?

12.6 Papers to Read Next

Bryant (1980) to see the closest formal predecessor; Jacklin (1987) to see the deposits-versus-equity qualification; Goldstein and Pauzner (2005) to see how equilibrium selection is eventually restored; and Diamond and Rajan (2001) to see the capital-structure extension this review’s Idea 4 builds on.

13. Conclusion

13.1 One-Sentence Thesis

A demand deposit contract is the efficient response to privately observed liquidity risk precisely because it pays whoever arrives first in full, and that same feature necessarily also supports a second, purely self-fulfilling equilibrium in which the bank fails for no reason connected to its underlying assets.

13.2 Three Most Important Insights

First, private information about an idiosyncratic risk can defeat even a complete, well-functioning market, which is why banking exists as an institution distinct from market trading. Second, risk-sharing value and fragility are joint products of the same contractual structure, not separable design choices. Third, not all remedies to fragility are equivalent: suspension of convertibility and deposit insurance coincide only in the absence of aggregate uncertainty, and diverge sharply once that uncertainty is present.

13.3 Strongest Evidence

The internal, constructive proof that the demand deposit contract implements the planner’s optimal allocation (Section 3.4) as one equilibrium, and the parallel, equally rigorous proof that a strictly worse equilibrium exists under identical primitives (Section 3.5) — together, these are airtight given the model’s assumptions, which is the strongest form of evidence a theory paper can offer.

13.4 Biggest Weakness

The complete absence of moral hazard: the paper’s policy conclusion in favor of government deposit insurance is derived in an environment where insuring the bank cannot possibly change the bank’s asset-side behavior, which is exactly the channel through which real-world deposit insurance has historically generated its most serious costs.

13.5 Most Important Research Gap

The absence of any equilibrium-selection mechanism — the model cannot say when a run is more or less likely, only that one is possible, leaving the paper’s policy relevance dependent on extensions (principally the global-games literature) that arrived two decades later.

13.6 Best Research Opportunity

Using fully observable, depositor-level blockchain data from stablecoin and DeFi lending-pool redemption events (Idea 1, Section 11) to separate panic-driven from information-driven runs with a precision no historical banking data set can offer.

13.7 Three Papers to Read Next

Jacklin (1987), “Demand Deposits, Trading Restrictions, and Risk Sharing”; Goldstein and Pauzner (2005), “Demand-Deposit Contracts and the Probability of Bank Runs”; and Gorton (1988), “Banking Panics and Business Cycles,” for the empirical counterpart this theoretical review deliberately leaves for a companion piece.

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